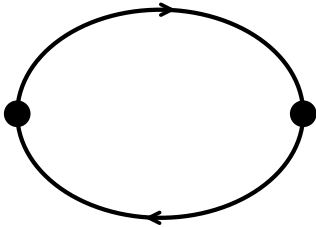
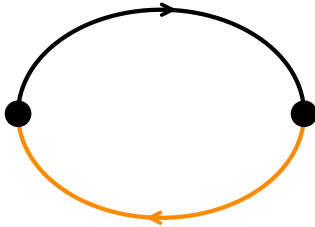


7 distinct 1-loop loop integrals at  $E = 3$  (matches thesis p.91-92, Eqs. 2.122-2.123) via rdft.crn

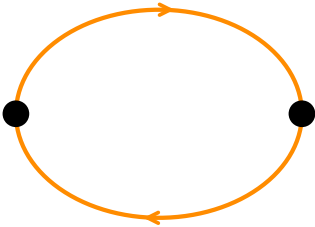
$V_{\text{branch}} + V_{\text{QC5}}$  int: AA  
ext:  $\bar{1}A\text{-out } 1B\text{-in } 1B\text{-out}$   $s(G) = 1$



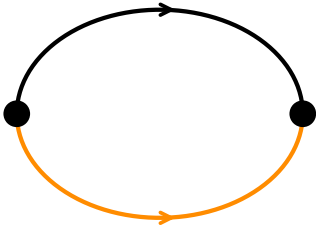
$V_{\text{QC1}} + V_{\text{QC4}}$  int: AB  
ext:  $1A\text{-out } 2B\text{-in}$   $s(G) = 1$



$V_{\text{QC2}} + V_{\text{QC3}}$  int: BB  
ext:  $2A\text{-out } 1B\text{-in}$   $s(G) = 1$

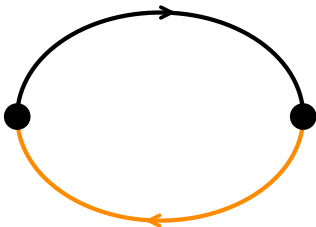


$V_{\text{QC2}} + V_{\text{QC5}}$  int: AB  
ext:  $1\bar{A}\text{-out } 1B\text{-in } 1B\text{-out}$   $s(G) = 1$



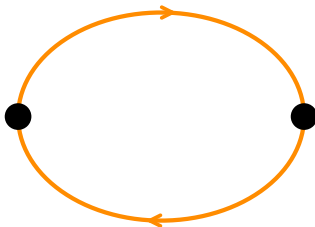
TRACE OBSERVABLE

$V_{\text{QC2}} + V_{\text{QC5}}$  int: AB  
ext:  $1\bar{A}\text{-out } 1B\text{-in } 1B\text{-out}$   $s(G) = 1$   
 $|\text{Aut}| = 1$ : no nontrivial automorphism



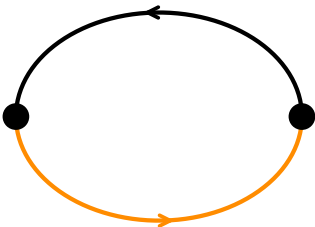
TRACE OBSERVABLE

$V_{\text{QC2}} + V_{\text{QC5}}$  int: BB  
ext:  $1A\text{-in } 2\bar{A}\text{-out}$   $s(G) = 1$   
 $|\text{Aut}| = 1$ : no nontrivial automorphism



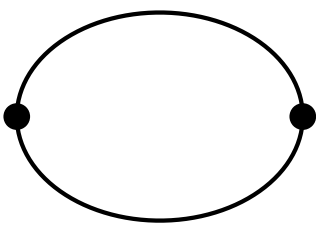
TRACE OBSERVABLE

$V_{\text{QC5}} + V_{\text{QC6}}$  int: AB  
ext:  $2A\text{-out } 1B\text{-in}$   $s(G) = 1$   
 $|\text{Aut}| = 1$ : no nontrivial automorphism



TRACE OBSERVABLE

$V_{\text{branch}} + V_{\text{branch}}$  int: AA  
ext:  $2A\text{-out}$   $s(G) = 2$  [thesis Eq. 3.25]  
 $|\text{Aut}| = 1$ : no nontrivial automorphism



TRACE OBSERVABLE

$|\text{Aut}| = 1$ : no nontrivial automorphism

RG-RELEVANT

$|\text{Aut}| = 1$ : no nontrivial automorphism

TRACE OBSERVABLE

$|\text{Aut}| = 1$ : no nontrivial automorphism

REFERENCE BUBBLE (E=2)

vertex-swap symmetry (same vertex type, lines invariant)